

Global Energy Storage

Global Energy Storage: How to power AI with solar-plus-storage



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The industry has long viewed solar-plus-storage as insufficient for baseload power particularly for 24/7 AI-driven demand. Masdar and EWE's gigascale solar-plus-storage project challenges this view with the world's first deployment of firm renewable power at scale. Our analysis leaves us incrementally more constructive on solar-plus-storage as a competitive source of baseload power, with Sungrow and CATL as key beneficiaries.

Solar-plus-storage can deliver baseload reliability at scale. UAE's Masdar and EWE are building 5.2 GW of solar combined with 19 GWh of storage (19 hours duration) to deliver ~1 GW of continuous power, with completion targeted by 2027. Based on our analysis, the system could achieve ~99.6% system reliability, representing a structural shift in renewables from intermittent energy sources to providers of firm capacity.

Economics are competitiveness particularly for higher gas price markets. While the upfront capex is high at ~\$6,000/kW, solar-plus-storage benefits from low operating costs and zero fuel exposure. At an estimated LCOE of \$97/MWh for the project, solar-plus-storage can compete with gas-fired power at gas price of ~\$8/mmbtu or higher. At 12-hour storage scenario, the LCOE falls to around \$80/MWh which can still achieve high system reliability of 95%. The attractiveness of solar-plus-storage as baseload is important given ongoing volatility and disruption in global gas supply. That said, we still believe gas-fired power is favored in regions with abundant low cost gas supply such as the US.

Faster deployment versus gas and nuclear is a key advantage. Solar and storage projects can be delivered in roughly two years, compared with current gas turbine lead times of around four years due to supply constraints, and even longer timelines for nuclear, typically six years or more.

The key constraint is access to high solar irradiance and land. While the system is replicable, we think it is still limited to solar-rich regions with abundant, low-cost land. This project alone requires ~60 km² of land—roughly the size of Manhattan—highlighting the significant physical footprint required to deliver firm renewable power at scale.

Storage is the primary driver of system economics rather than solar. ESS account for roughly half of total project capex, meaning cost competitiveness is primarily driven by storage cost, efficiency, and performance rather than module pricing.

We expect global ESS demand to grow at ~34% CAGR over the next five years, driven by the need for firming renewable generation and grid stability. Within the value chain, **CATL** leads in ESS battery supply and technology, while **Sungrow** is a key player in system integration, inverters, and broader power conversion solutions, positioning both as primary beneficiaries of this structural shift.

BERNSTEIN TICKER TABLE

Ticker	Rating	12 Jun 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
300750.CH (CATL)	O	CNY	394.85	800.00	18.9%	CNY	16.14	21.95	28.77	24.5	18.0	13.7
247540.KS (EcoPro BM)	U	KRW	167,800	140,000	32.9%	KRW	403.00	854.00	1,963.00	416.4	196.5	85.5
051910.KS (LG Chem)	M	KRW	320,000	298,000	20.7%	KRW	13,258.70	2,042.47	31,118	(24.1)	156.7	10.3
373220.KS (LGES)	M	KRW	387,000	347,000	(9.9)%	KRW	(5,308.10)	1,811.04	9,452.97	(72.9)	213.7	40.9
003670.KS (Posco Future M)	U	KRW	188,800	190,000	13.4%	KRW	(2,740.96)	376.72	898.20	(68.9)	501.2	210.2
006400.KS (SDI)	M	KRW	507,000	520,000	168.1%	KRW	(9,933.80)	2,099.51	18,376	(51.0)	241.5	27.6
300274.CH (Sungrow)	O	RMB	148.76	185.00	90.9%	RMB	6.55	8.22	9.33	22.7	18.1	15.9
002466.CH (Tianqi Lithium)	O	CNY	62.50	73.00	63.3%	CNY	0.28	3.29	6.18	221.5	19.0	10.1
9696.HK (Tianqi Lithium)	O	HKD	47.58	61.00	36.7%	CNY	0.28	3.29	6.18	145.5	12.5	6.6
034020.KS (Doosan Enerbility)	O	KRW	90,100	95,000	32.8%	KRW	173.88	45.91	687.53	518.2	N/M	131.0
3750.HK (CATL)	M	HKD	670.00	770.00	77.7%	CNY	16.14	21.95	28.77	35.8	26.3	20.1
ASIAx			1,966.72									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

003670.KS, 034020.KS base year is 2024;

Source: Bloomberg, Bernstein estimates and analysis.

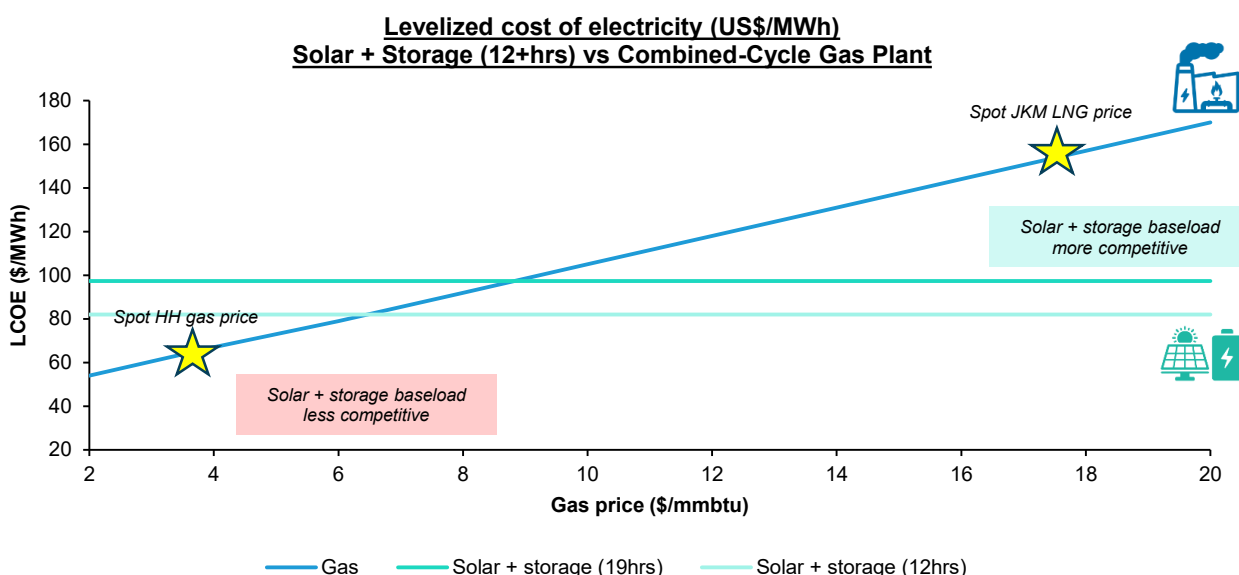
INVESTMENT IMPLICATIONS

In a structurally power-constrained world, solar-plus-storage is emerging as an increasingly important solution to deliver stable, baseload-like power, particularly for fast-growing AI and data center demand. The Masdar project demonstrates that, with sufficient overbuild and long-duration storage, renewables could achieve high system reliability while offering competitive economics in higher gas price environments, alongside the key advantages of being emissions-free and having no fuel price risk. While scalability remains dependent on geography, land availability, and grid infrastructure, the overall direction is clearly positive, with costs expected to decline further through scale and ongoing technology improvements, including next-generation batteries such as sodium-ion. As storage now drives system economics and demand continues to accelerate, we see strong upside for global ESS deployment and recommend Sungrow and CATL as key beneficiaries of this structural shift.

DETAILS

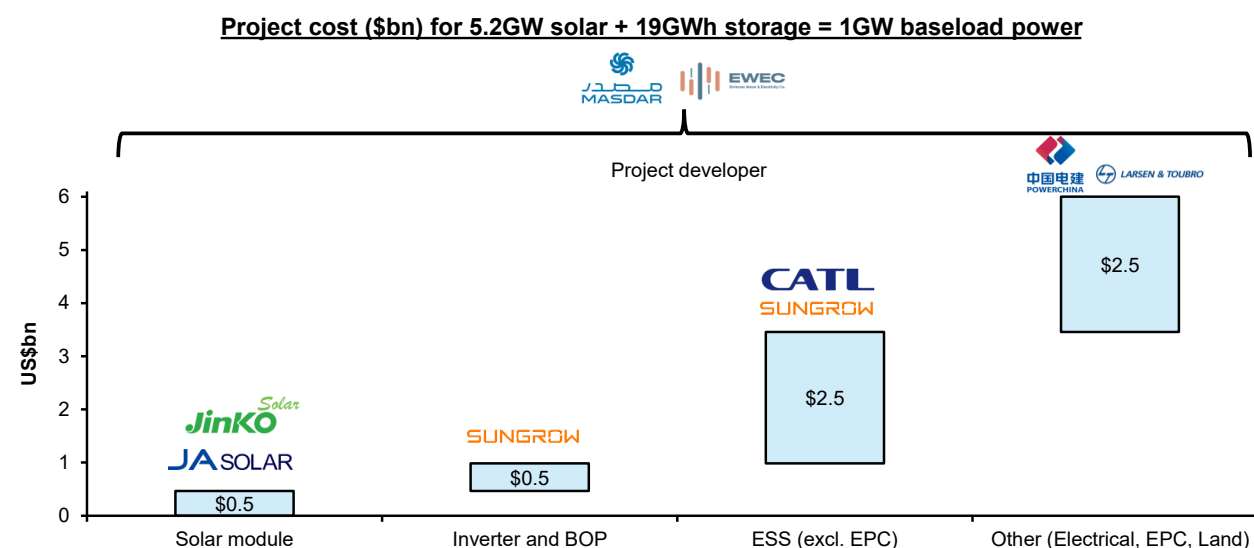
The UAE's Masdar and EWEC project demonstrates at scale that solar-plus-storage can deliver firm baseload power. This fundamentally reframes the role of renewables: the key question is no longer whether solar can generate cheap energy, but whether it can reliably meet a continuous load profile when paired with sufficient storage and overbuild. The project combines 5.2 GW of solar with 19 GWh of storage to deliver ~1 GW of continuous power, with completion targeted by 2027. This represents the first gigascale attempt to overcome solar intermittency through system design, and therefore serves as a key proof point for the industry. We find that solar-plus-storage is increasingly cost competitive in higher gas price environments. At current cost levels, it can compete with gas-fired generation when fuel prices are elevated, making it an attractive alternative amid volatility in global gas markets. However, gas remains structurally advantaged in regions with persistently low fuel costs, such as the US.

EXHIBIT 1: **Solar plus storage (12+ hours) are competitive against gas-fired power at high gas prices**



Source: Company data, Bernstein analysis

We estimate total project capex at approximately \$6bn for 1GW of firm solar-plus-storage capacity (or ~\$6,000/kW). This can be broadly broken down into ~\$0.5bn for solar modules, ~\$0.5bn for inverters and balance-of-plant, and ~\$2.5bn for ESS, with the remaining ~\$2.5bn covering grid/electrical infrastructure, EPC, land, and other development costs. The key takeaway is that while solar generation itself is already low cost, storage dominates the economics, accounting for roughly half of total system cost. As a result, the competitiveness of baseload solar-plus-storage is primarily driven by battery cost declines rather than further reductions in module pricing. This reinforces our view that continued cost reductions in ESS will be the critical enabler for scaling firm renewable power.

EXHIBIT 2: We estimate total project capex at approximately \$6bn (or ~\$6,000/kW). Energy storage dominates the economics


CATL and Sungrow are covered by Bernstein. All other companies are not covered by Bernstein.

Source: Company data, Bernstein analysis and estimates

Solar-plus-storage is emerging as a credible alternative to traditional baseload generation, offering several advantages over gas and nuclear, but with clear trade-offs. The Masdar project demonstrates that, with sufficient overbuild and long-duration storage, renewables can achieve near-baseload reliability (~99% uptime), while also benefiting from faster deployment timelines of ~2 years compared to ~3–6 years for gas (given turbine bottlenecks) and ~6+ years for nuclear. Economically, solar-plus-storage is increasingly competitive in higher gas price environments, providing cost certainty without fuel price exposure. However, it remains constrained by geography, requiring high solar irradiance and large land footprints, and its economics are heavily dependent on battery costs, which account for ~50% of capex. By contrast, gas remains more flexible and competitive in low-cost fuel markets such as the US, while nuclear offers the highest level of clean baseload reliability but with significantly longer build times and higher execution risk. As such, solar-plus-storage can deliver very competitive firm power in select regions, particularly for new demand such as AI and data centers.

EXHIBIT 3: Comparison of solar-plus-storage, gas and nuclear power for baseload power

Metric	Solar + Storage (12+ hours)	Gas - CCGT	Nuclear
Uptime	~90–99.x% ✓	~95–99.9% ✓	~95–99.9% ✓
LCOE	~\$80–100/MWh ✓	~\$50/MWh (\$3.5/mmbtu) ~\$110/MWh (\$12/mmbtu) ✓	~\$50–150/MWh
Lead time	~2 years ✓	~3–6 years	6+ years
Land requirement	~60 km ²	~0.02–0.05 km ² ✓	~1–4 km ² ✓
Carbon emissions	Zero ✓	~0.35–0.4 tCO ₂ /MWh	Near zero ✓
Fuel risk	none ✓	gas - high fuel cost	uranium - low fuel cost ✓
Bottleneck	location, grid connection	turbine supply, gas supply	long development

Assuming a build for 1GW of baseload power

Source: Company data, Bernstein analysis

SOLAR-PLUS-STORAGE PROVIDES BASELOAD POWER

The UAE's Masdar and EWEC project represents a step-change in solar-plus-storage system design. Rather than optimizing for peak shaving, it is engineered to deliver round-the-clock baseload power. By pairing 5.2 GW of solar with 19 GWh of storage, ~1 GW of continuous power output can be achieved to match stable power load of 1GW. This “abundance” is critical: excess midday generation is not curtailed but stored in the battery system (equivalent to ~19 hours of duration) and discharged through

the evening and overnight. In effect, the project replaces curtailment with energy shifting, reshaping variable solar generation into a stable, near-continuous output profile.

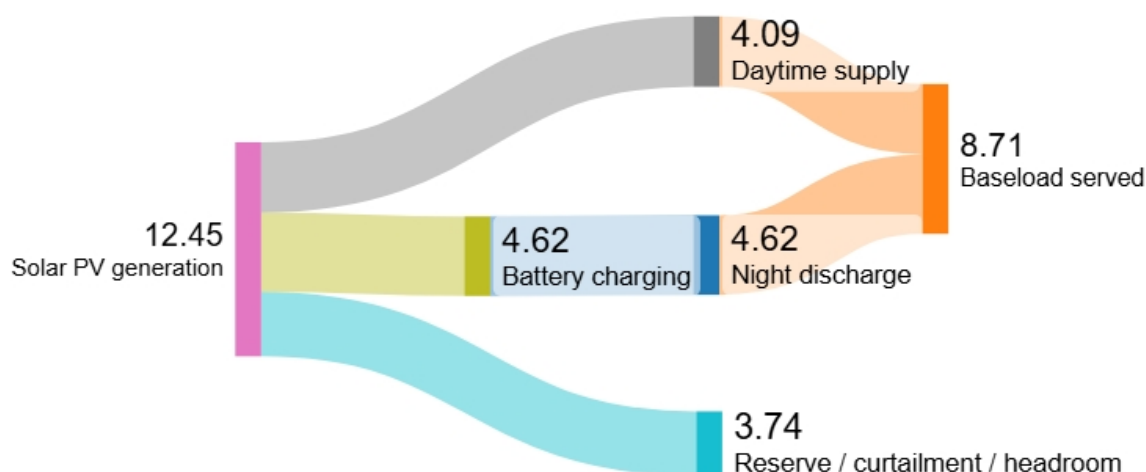
EXHIBIT 4: By pairing 5.2 GW of solar with 19 GWh of storage, 1 GW of continuous power output can be achieved to match stable a power load of 1GW



Source: Company presentation

How does this work? With 5.2 GW of installed solar capacity, the project could generate approximately **12.45 TWh of annual solar output**, implying a ~27% capacity factor based on the local irradiance profile. This equates to ~34 GWh per day of energy production vs. a constant 24 GWh/day load for 1 GW baseload, highlighting that the system is intentionally **overbuilt by ~40% on an energy basis**. Of this total generation, around 4.09 TWh is used to directly supply daytime demand, while ~4.62 TWh is stored in the battery during the day and discharged at nighttime to maintain continuous output. The remaining ~3.74 TWh represents surplus generation, which serves as headroom to meet storage losses, seasonal variability, and system reliability requirements. With this system, we estimate a **99.6% system reliability can be achieved**.

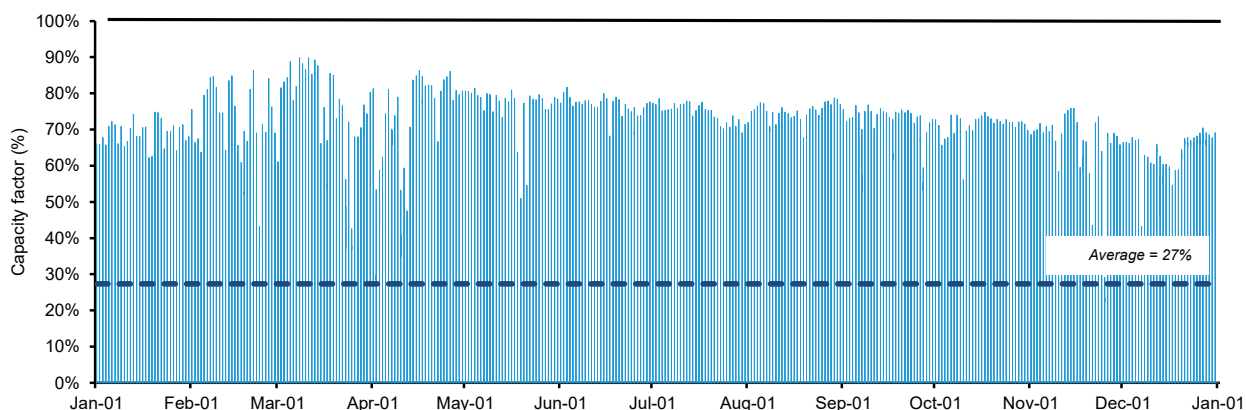
EXHIBIT 5: Annual power output of the project (TWh). Solar and storage overbuilt enables continuous 1 GW output



Source: Company data, Bernstein analysis

We have carried out an analysis of what the operations and economics would look like for the project based on the region's irradiance profile and the system design. Building solar capacity in UAE offers significant advantages due to the region's abundant solar resources, characterized by high solar irradiance and vast available land, enabling large-scale, cost-effective solar power generation. Additionally, favorable government policy and expanding transmission infrastructure support export of electricity. Based on the irradiance profile, we expect the project can generate 12.45TWh of electricity generation with a 1-axis tilt and azimuth. On average, the annual capacity factor is approximately 27% assuming no curtailment.

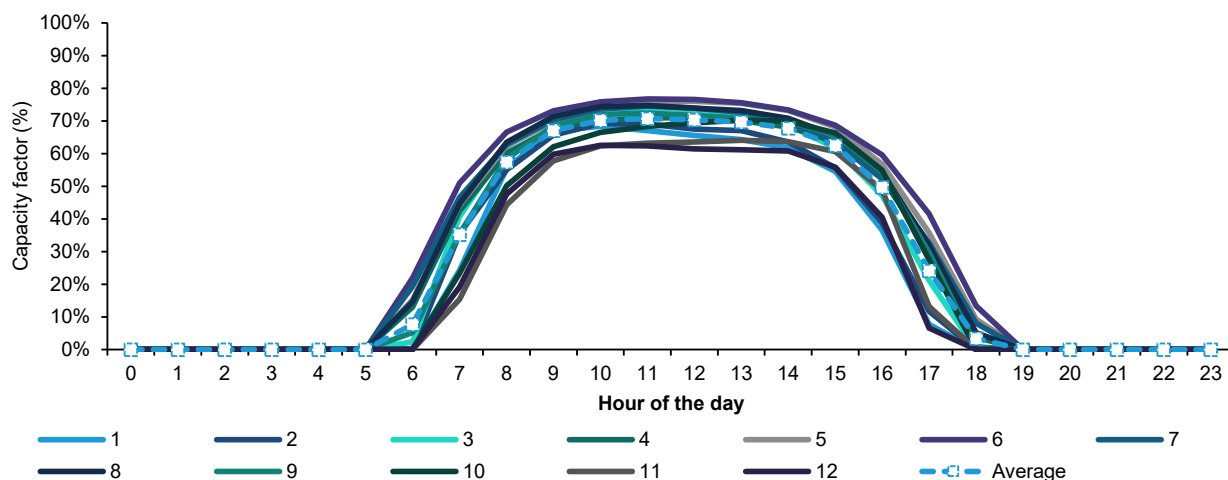
EXHIBIT 6: We have modeled 5.2GW of solar capacity. The solar farm can achieve an average capacity factor of 27% although we have not factored in curtailment and power load



Source: Company data, Bernstein analysis

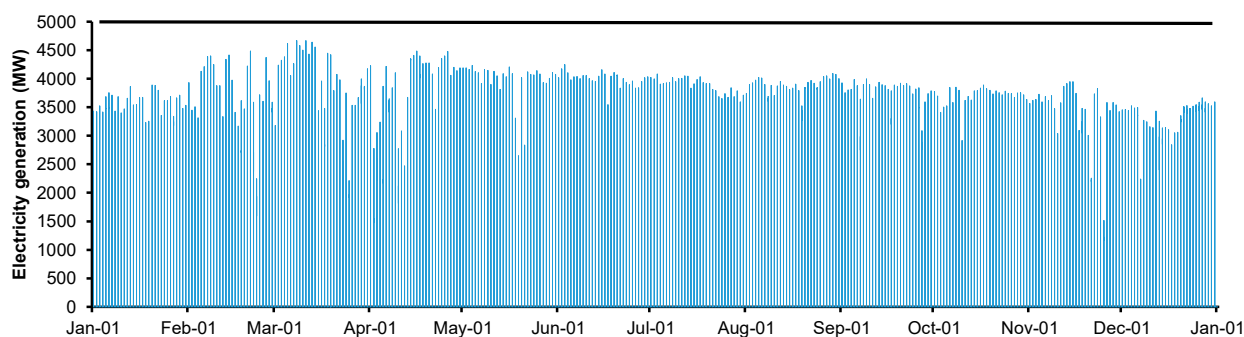
Solar has significant variability in capacity factor by months and hours of the day. Capacity factors are generally higher in summer months with longer daylight hours and stronger solar irradiance. Solar power generation are typically between 6am to 5pm with peak generation between 10am to 1pm where the sun's ray strikes the panels most directly.

EXHIBIT 7: Solar power is mostly generated between 7am to 5pm, which means there is no power for more than 12 hours of the day without storage or other power source



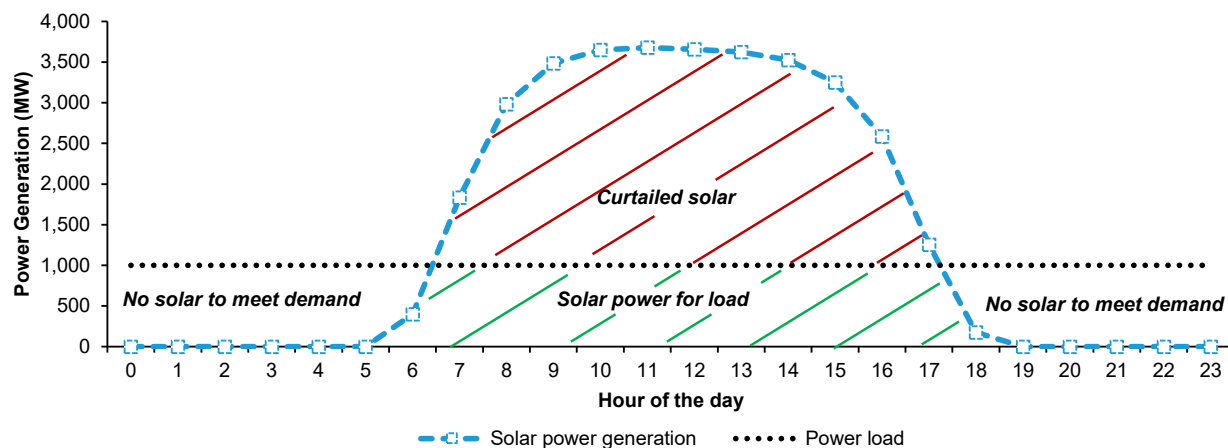
Source: Company data, Bernstein analysis

The hourly generation profile below illustrates significant volatility in electricity generation without storage. Power generation drops to zero during peak evening hours. For a 5.2GW solar farm, the solar project can generate 12.45TWh of electricity annually.

EXHIBIT 8: Annually, the project could generate 12.45TWh of electricity, although this is interruptible

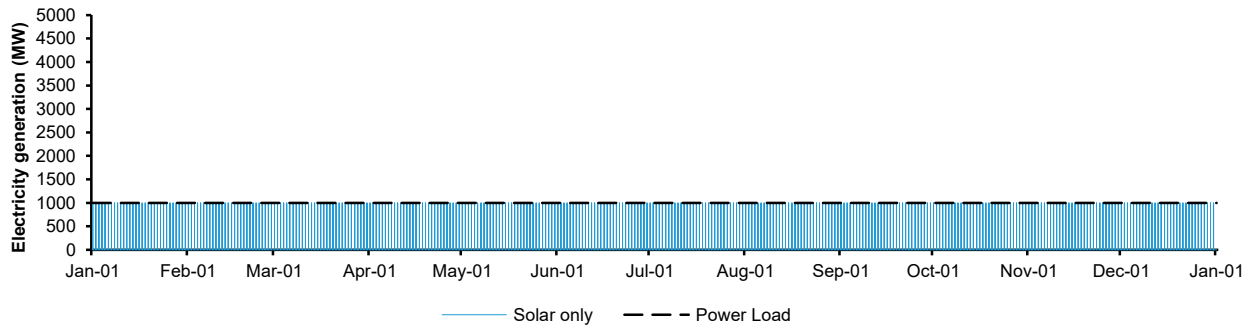
Source: Company data, Bernstein analysis

While the solar project can theoretically achieve 27% capacity factor, realistically utilization is lower due to curtailment driven by mismatch between supply and demand. After factoring a continuous power load of 1GW, solar power generation would be curtailed at peak production around midday, as supply exceeds the demand. This mismatch leads to intentional reduction of solar output to maintain grid stability and prevent overloads. Curtailment results from limited grid flexibility and insufficient storage.

EXHIBIT 9: While the solar project can theoretically achieve 27% capacity factor, realistically utilization is lower due to curtailment during midday without storage

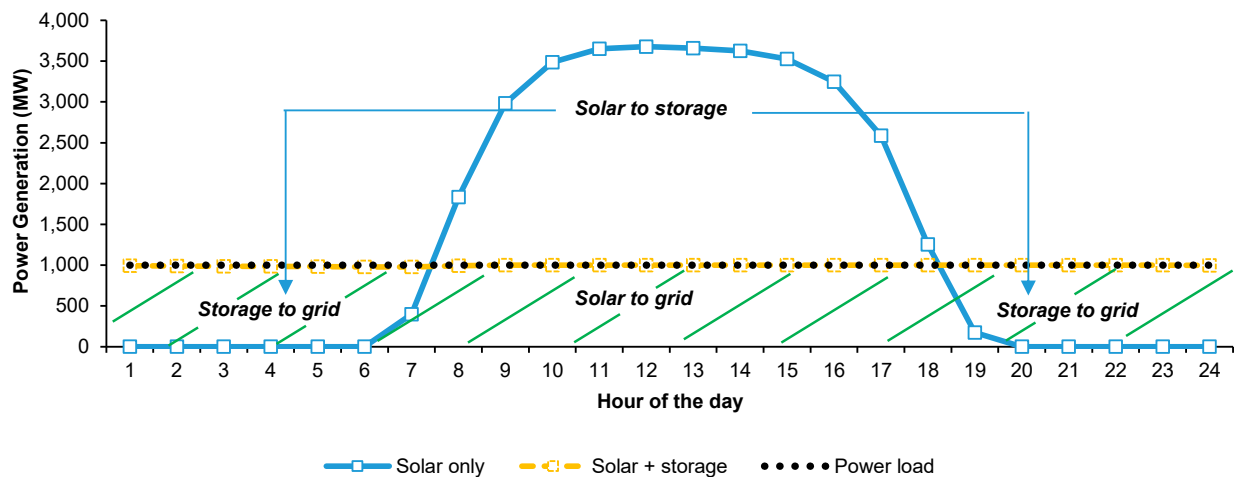
Source: Company data, Bernstein analysis

After factoring in curtailment, the actual solar power generation from the project is effectively reduced to match the power load during midday. As a result, the generation profile shows lower output during periods when the grid cannot absorb all the solar energy produced. This means the net generation curve is clipped or flattened at these times. Consequently, the curtailment results in lower project revenue and efficiency metrics. After factoring in the curtailment, capacity factor for the system falls to 9%.

EXHIBIT 10: **Without storage, the curtailment results in capacity factor falling to 9%**

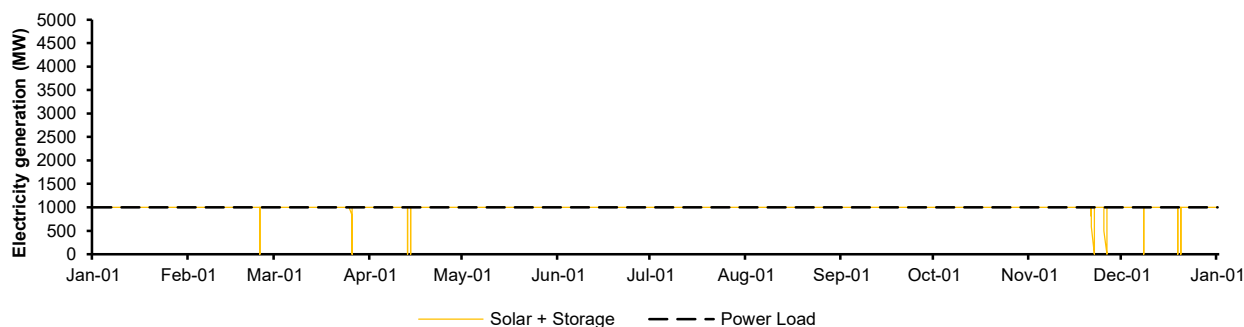
Source: Company data, Bernstein analysis

Batteries help to store excess power and discharge during higher power load in the morning and evening. Sizing battery storage alongside solar requires a detailed analysis of solar generation and load demand profiles to optimize system performance and cost-efficiency. Key factors include understanding the timing and magnitude of solar output and electricity consumption, determining the required storage duration for balancing supply and demand, and accounting for battery characteristics such as depth of discharge and efficiency. The sizing strategy aims to maximize consumption and ensure reliability during periods of low solar generation. A well-established system prevents power shortfalls and excessive capital investment, enhancing the overall value proposition of solar-plus-storage projects.

EXHIBIT 11: **1GW of continuous power output enabled by the project**

Source: Company data, Bernstein analysis

A key design feature of this project is its ~19 GWh of storage, which equates to ~19 hours of duration at a 1 GW output level. This is sufficient to cover essentially the entire non-solar generation window, including evening, overnight, and early morning hours when PV output is near zero. In practical terms, the system only needs to rely on real-time solar generation for a limited portion of the day, with storage providing the majority of off-peak supply. This long-duration capability is what enables the system to deliver near-continuous output, as it can bridge the full diurnal gap between production and demand. Importantly, the 19-hour duration is not excessive but deliberately allows for full night-time coverage while also providing buffer for efficiency losses and variability in solar generation. Combined with the ~40% energy overbuild, this ensures that sufficient energy can be stored each day to reliably sustain baseload delivery, even under less-than-ideal conditions. Given the irradiance profile and system build, continuous power output at 1GW is achieved almost entirely through the year.

EXHIBIT 12: **System sustains ~1 GW of near-continuous output throughout the year**

Source: Company data, Bernstein analysis

Below we summarize the key operating characteristics for different configuration of solar+storage systems at different duration. For 19-hour storage, the system is able to sustain ~1 GW of near-continuous output with ~99.7% uptime based on our modelling.

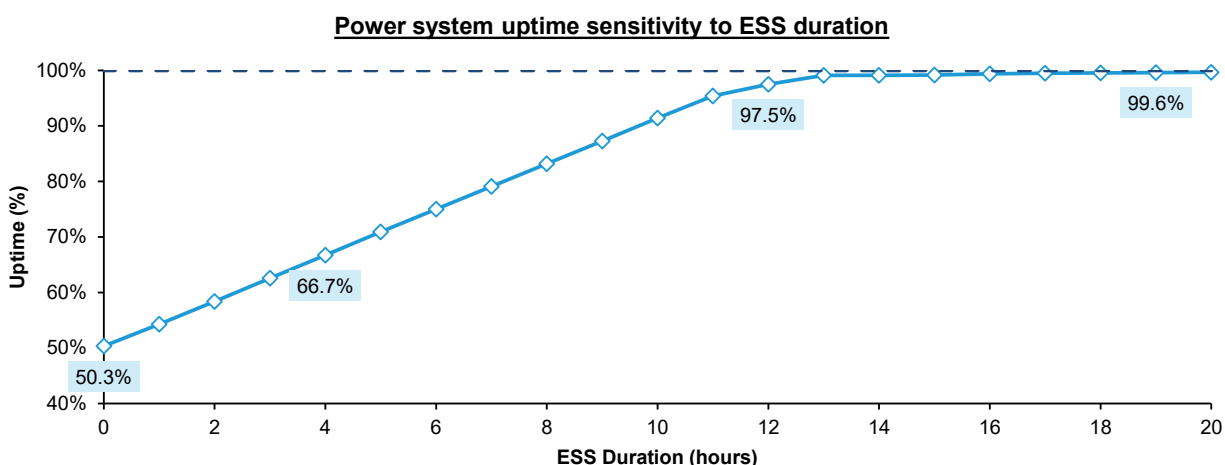
EXHIBIT 13: **Key operating metrics for solar vs solar-plus-storage**

		<i>Solar only</i>	<i>Solar + 4-hr ESS</i>	<i>Solar + 12-hr ESS</i>	<i>Solar + 19-hr ESS</i>
<u>Combined</u>					
<i>Daily power load</i>	<i>MW</i>	24,000	24,000	24,000	24,000
<i>Daily generation</i>	<i>MW</i>	11,217	15,215	22,933	23,879
<i>Uptime</i>	<i>%</i>	50.3%	66.7%	97.5%	99.6%
<u>Solar</u>					
<i>Capacity</i>	<i>MW</i>	5200	5200	5200	5200
<i>Effective utilization</i>	<i>%</i>	9%	12%	18%	19%
<u>Battery</u>					
<i>Battery size</i>	<i>MWh</i>	0	4,000	12,000	19,000
<i>Duration</i>	<i>hours</i>	0	4.0	12.0	19.0
<i>DoD per day</i>	<i>%</i>	0%	100%	98%	67%

Source: Company data, Bernstein analysis

Based on our modelling, this configuration can achieve ~99.7% uptime, effectively delivering baseload-like reliability. Even with a reduced configuration of ~12 hours of storage, uptime could still reach ~97.5%, highlighting the critical role of storage duration in determining system reliability. While short-duration systems (e.g., 2–4 hours) can improve utilization and reduce curtailment, they fully eliminate supply gaps, whereas longer duration storage (12+ hours) enables full day–night energy shifting. Together with deliberate solar overbuild, excess midday generation is stored rather than curtailed and discharged overnight, reshaping variable solar output into a flat, baseload-like profile, representing a structural shift from intermittent renewables to firm power.

EXHIBIT 14: Based on our modelling, this configuration (19 hours) can achieve ~99.7% uptime, effectively delivering baseload-like reliability

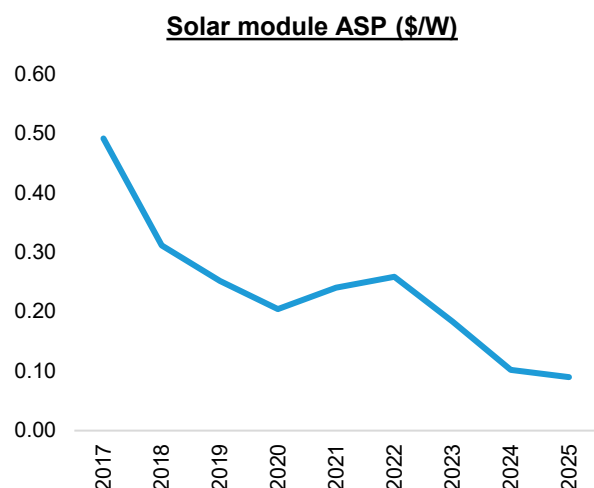


Source: Company data, Bernstein analysis

HOW MUCH WILL IT COST?

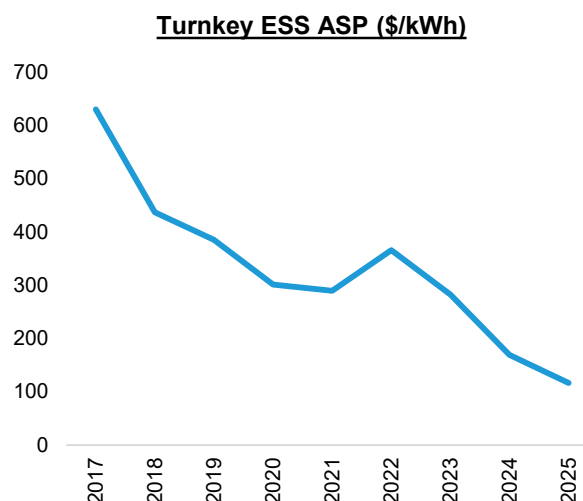
Solar and storage costs have declined significantly in recent years, making baseload solar-plus-storage systems increasingly viable. This reflects a combination of scale-driven manufacturing efficiencies, supply chain maturation, and rapid technology improvements across both solar and battery value chains. Solar modules have benefited from continuous efficiency gains, larger wafer formats, and aggressive capacity expansion in China, driving pricing to cyclical lows. At the same time, battery costs have declined with improvements in cell chemistry, energy density, and manufacturing scale, supported by strong growth in EV and stationary storage demand. As a result, prices are now at decade lows, with solar modules at around \$0.09/W, inverters at roughly \$0.09/W, and ESS costs at approximately \$130/kWh, materially improving system economics relative to historical levels and enabling new use cases such as firm power generation.

EXHIBIT 15: Solar ASP are at decade low which makes economics attractive



Source: Bloomberg, Bernstein analysis

EXHIBIT 16: ESS costs have also continued to decline in recent years

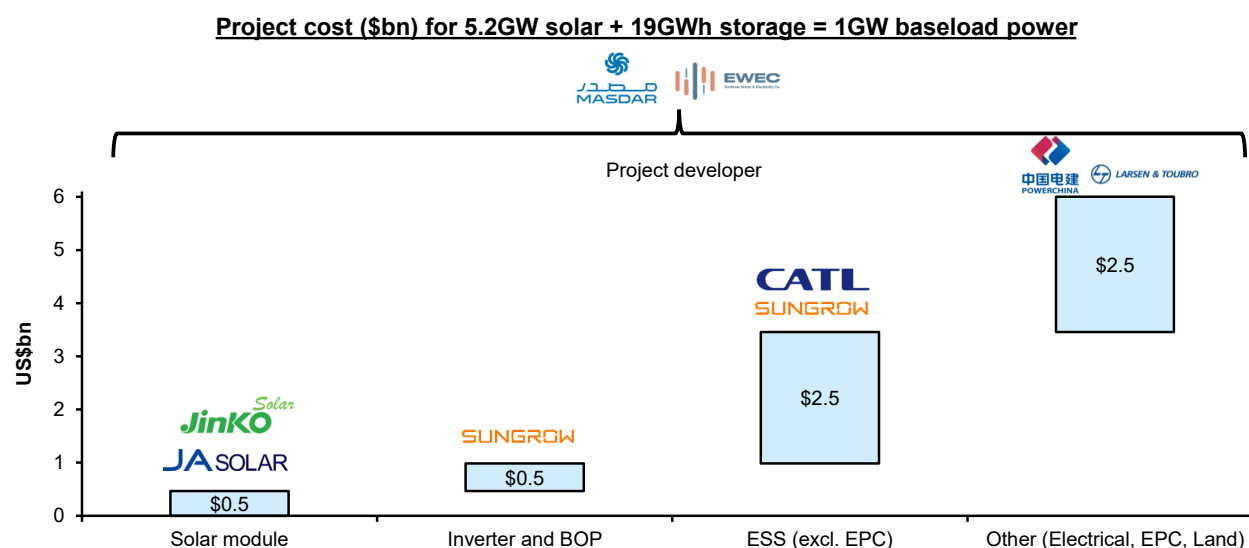


Source: Bloomberg, Bernstein analysis

We estimate total project capex at approximately \$6bn for 1 GW of firm solar-plus-storage capacity (or ~\$6,000/kW). This can be broadly broken down into ~\$0.5bn for solar modules, ~\$0.5bn for inverters and balance-of-plant, and ~\$2.5bn for ESS,

with the remaining ~\$2.5bn covering grid/electrical infrastructure, EPC, land, and other development costs. The key takeaway is that while solar generation itself is already low cost, storage dominates the economics, accounting for roughly half of total system cost. As a result, the competitiveness of baseload solar-plus-storage is primarily driven by battery cost declines rather than further reductions in module pricing. This reinforces our view that continued cost reductions in ESS will be the critical enabler for scaling firm renewable power.

EXHIBIT 17: We estimate total project capex at approximately \$6bn (or ~\$6,000/kW). Energy storage dominates the economics



CATL and Sungrow are covered by Bernstein. All other companies are not covered by Bernstein.

Source: Company data, Bernstein analysis and estimates

Below we summarize our key assumptions for the project. Based on our modelling, we estimate an LCOE of approximately \$97/MWh, assuming an 8% IRR.

EXHIBIT 18: Our assumptions for the solar + storage (19 hours) results in LCOE of \$97/MWh

Solar Assumptions	Units	Value
Capacity	MW	5,200
Capacity factor	%	10%
Electricity generation	GWh	4,555
Capital intensity	\$/kW	550
Capital cost	\$M	2,860
Fixed O&M cost	\$/kW/year	10.0

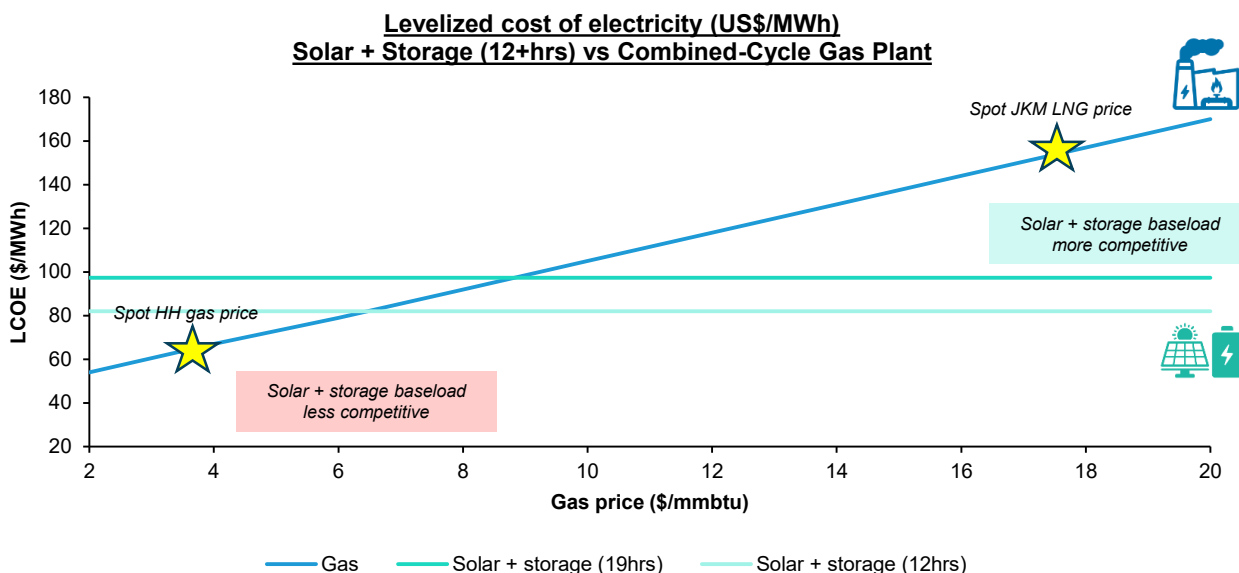
Energy Storage Assumptions	Units	Value
Energy Storage	MWh	19,000
Capital intensity	\$/kWh	165
Capital cost	\$M	3,135
DoD Cycles/Day	%	67%
Annual Degradation	%	3%
Efficiency	%	90%
Number of cycles	Times	2446

Economics	Units	Value
IRR	%	8%
LCOE	\$/MWh	97

Source: Company data, Bernstein analysis

Economics are competitiveness in higher gas price environments. At an estimated LCOE of \$97/MWh for the Masdar project, and around \$80/MWh under a shorter-duration 12-hour storage scenario, solar-plus-storage can compete with gas-fired power at gas price of ~\$8/mmbtu or higher. This is particularly relevant given ongoing volatility and disruption in global gas supply. That said, gas remains favored in regions with structurally low fuel costs such as the US.

EXHIBIT 19: Solar plus storage (12+ hours) are competitive against gas-fired power at high gas prices



Source: Company data, Bernstein analysis

GLOBAL POWER AND ESS DEMAND OUTLOOK

Since the beginning of the year, we have adopted a more positive outlook on ESS and significantly elevated our forecast to capture the higher ESS penetration driven by improved economics and the critical role of ESS in ensuring grid stability. We expect global ESS demand to grow at ~34% CAGR over the next five years, driven by the need for firming renewable generation and grid stability. Within the value chain, **CATL** leads in ESS battery supply, while **Sungrow** is a key player in system integration, inverters, and broader power conversion solutions, positioning both as primary beneficiaries of this structural shift.

EXHIBIT 20: **Global power and ESS demand outlook**

	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2050E
Total power capacity (GW)	7,690	8,053	8,495	9,160	9,973	10,812	11,755	12,799	13,887	15,012	16,170	40,260
Total solar and wind capacity (GW)	1,531	1,802	2,127	2,686	3,399	4,144	4,978	5,877	6,861	7,878	8,920	33,377
Share of solar and wind (%)	20%	22%	25%	29%	34%	38%	42%	46%	49%	52%	55%	83%
Net added solar and wind capacity (GW)	262	271	324	560	713	745	834	898	984	1,017	1,042	1,403
y-o-y (%)		3%	20%	73%	27%	4%	12%	8%	10%	3%	2%	1%
Cumulative ESS capacity (GW)	18	27	45	90	165	281	469	773	1,169	1,576	1,991	9,755
As of total capacity (%)	0%	0%	1%	1%	2%	3%	4%	6%	8%	11%	12%	24%
As of solar and wind capacity (%)	1%	2%	2%	3%	5%	7%	9%	13%	17%	20%	22%	29%
Net added ESS capacity (GW)	6	10	18	45	75	116	189	304	396	408	414	410
y-o-y (%)		63%	82%	154%	68%	55%	62%	61%	30%	3%	2%	2%
Duration (hours)	2.0	2.1	2.1	2.1	2.3	2.5	2.6	2.8	3.1	3.3	3.4	4.1
Cumulative ESS capacity (GWh)	36	58	95	191	375	700	1,228	2,200	3,624	5,173	6,830	39,789
Annual battery installation (GWh)	12	22	37	96	184	325	542	1,009	1,501	1,712	1,915	4,406
y-o-y	61%	78%	71%	159%	91%	77%	67%	86%	49%	14%	12%	3%
Annual ESS battery shipment/demand (GWh)	21	44	121	185	301	550	1,023	1,523	1,737	1,943	2,109	4,605
y-o-y	73%	114%	175%	53%	63%	83%	86%	49%	14%	12%	9%	3%

Source: Government websites, Bloomberg, InfoLink, Bernstein analysis and estimates

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

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